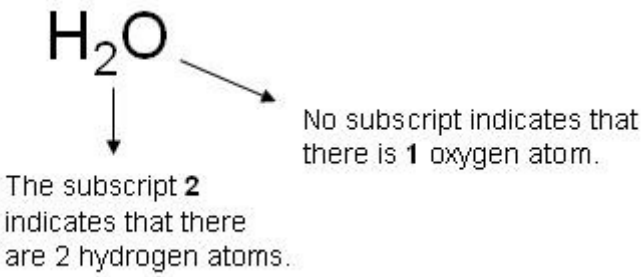
Balancing Chemical Equations

Remember that with a formula you are shown how many of each atom is in one molecule of a compound

The **subscript** is the name given to the number directly to the right of the element. It tells you how many atoms there are of that particular element. If there isn't a subscript, that means there is only one atom. For example, let's look at the water molecule.



Reactants=the chemicals at the start of a chemical reaction.

Products=the chemicals made in a chemical reaction.

The reactants are on one side of the arrow.

Products are on the other side of the arrow.

A balanced chemical equation has equal numbers of all elements on each side of an equation.

Some equations are already balanced for example. Carbon burning in oxygen.

Reactants **products**

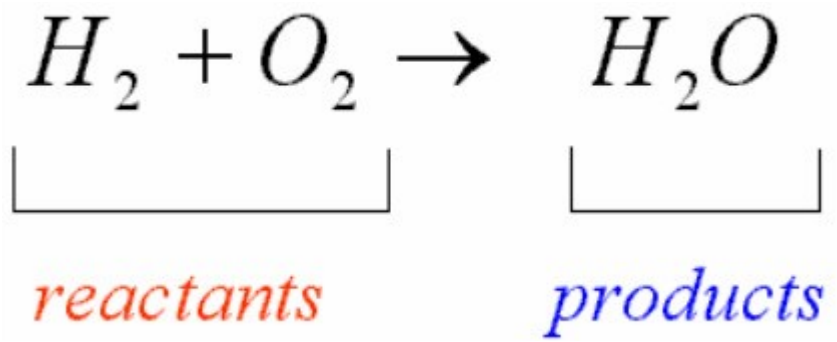
Carbon + oxygen gas-----carbon dioxide

$C + O_2$ ---------- CO_2

1 carbon 1 carbon

2 oxygen 2 oxygen

Some do not

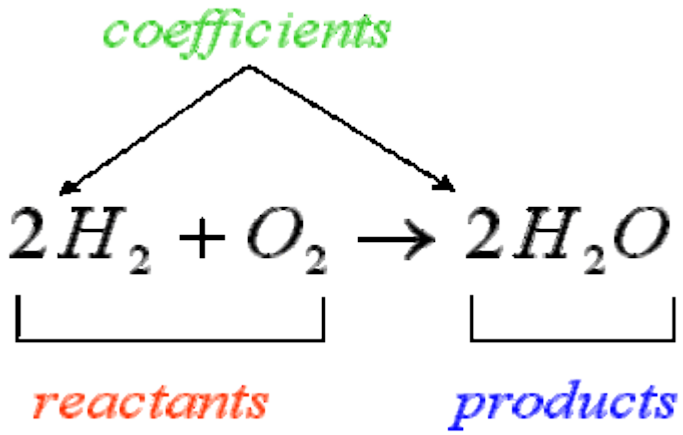


2 hydrogen | 2 hydrogen

2 oxygen | ***1 oxygen***

Notice there are more oxygens on the reactants side than on the products side.

To balance this equation we will need the following



4 hydrogen | 4 hydrogen

2 oxygen | ***2 oxygen***

The **coefficient** is the number that goes in front of a chemical formula